AstraQuantum Tech

Summer of Cybersecurity 2026

WEEK 03 — NETWORKING FOR CYBERSECURITY

Assessment Report

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Assessment	Week 3: Networking for Cybersecurity
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Submission Date	25 August 2026
Submission Platform	Google Classroom

Project Links

GitHub Repository: github.com/UICipher/week3-networking-cybersecurity

LinkedIn Post: lnkd.in/p/dT57gvPd

Portfolio Links	
GitHub Repo	https://github.com/UICipher/week3-networking-cybersecurity/
LinkedIn Post	https://lnkd.in/p/dT57gvPd

The repo contains the full .pkt file plus every screenshot referenced in this report, organized with a README covering the topology, addressing, and lab results. The LinkedIn post covers the same work in a shorter form for a general audience.

1. Lab Objectives

This week's assignment was mainly about getting hands-on with networking instead of just reading about it, and connecting that to how a security analyst would actually look at a network. Over the course of the lab I had to:

- Design and build a small enterprise-style network using Cisco Packet Tracer, consisting of three routers, three switches, three LANs, and three end devices.
- Configure static routing between all routers so that every LAN could reach every other LAN.
- Verify connectivity using ping, traceroute, and routing-table inspection.
- Use Packet Tracer's Simulation Mode to capture and analyze ARP, ICMP, and TCP/UDP traffic at the packet level.
- Connect networking concepts (MAC addressing, IP addressing, routing, ports) to real security-monitoring use cases.
- Intentionally break the network and practice systematic troubleshooting.
- Complete supporting online labs on LetsDefend and TryHackMe to reinforce packet analysis and network-service concepts.

The overall flow I followed was: Build → Configure → Test → Analyze → Secure, which is basically the same order a junior network/SOC analyst would work through in a real job.

2. Network Topology

The final topology built in Cisco Packet Tracer consists of three routers (R1, R2, R3) connected in a chain, each with its own LAN and switch, and two point-to-point router links.

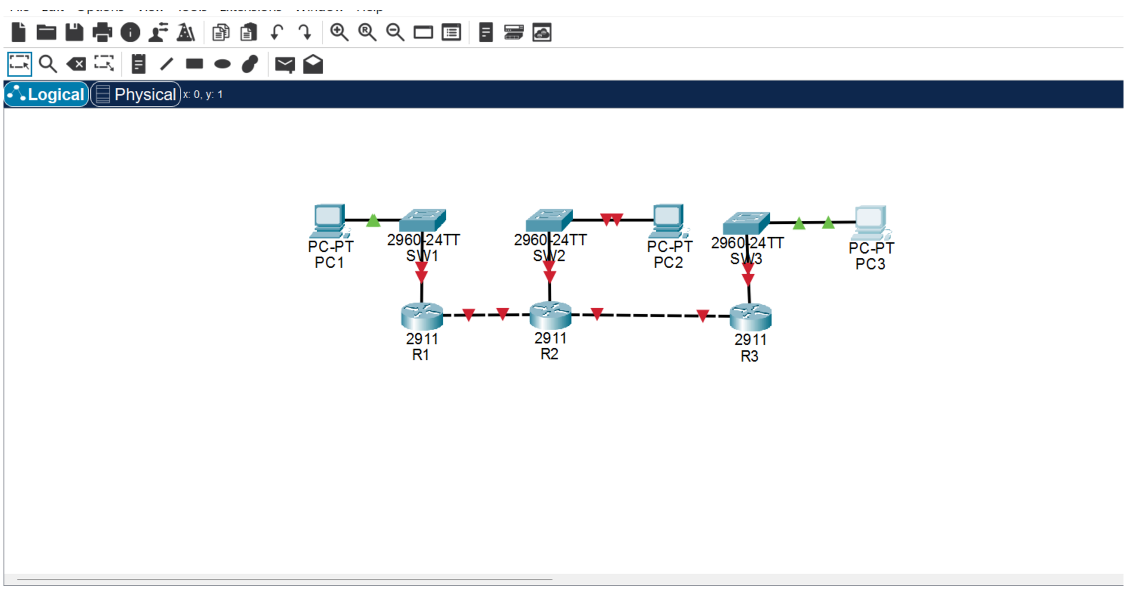


Figure 1 — Completed three-router topology: PC1–SW1–R1, R2 with LAN2/PC2, R3–SW3–PC3.

3. IP Addressing

The addressing plan below follows the scheme specified in the assignment exactly, using /24 subnets for each LAN and /30 subnets for the router-to-router links.

Segment	Device	Interface	IP Address	Subnet Mask
LAN 1	R1	G0/0	192.168.10.1	255.255.255.0
LAN 1	PC1	NIC	192.168.10.10	255.255.255.0
R1 ↔ R2	R1	G0/1	10.0.12.1	255.255.255.252
R1 ↔ R2	R2	G0/0	10.0.12.2	255.255.255.252
LAN 2	R2	G0/1	192.168.20.1	255.255.255.0
LAN 2	PC2	NIC	192.168.20.10	255.255.255.0
R2 ↔ R3	R2	G0/2	10.0.23.1	255.255.255.252
R2 ↔ R3	R3	G0/0	10.0.23.2	255.255.255.252
LAN 3	R3	G0/1	192.168.30.1	255.255.255.0
LAN 3	PC3	NIC	192.168.30.10	255.255.255.0

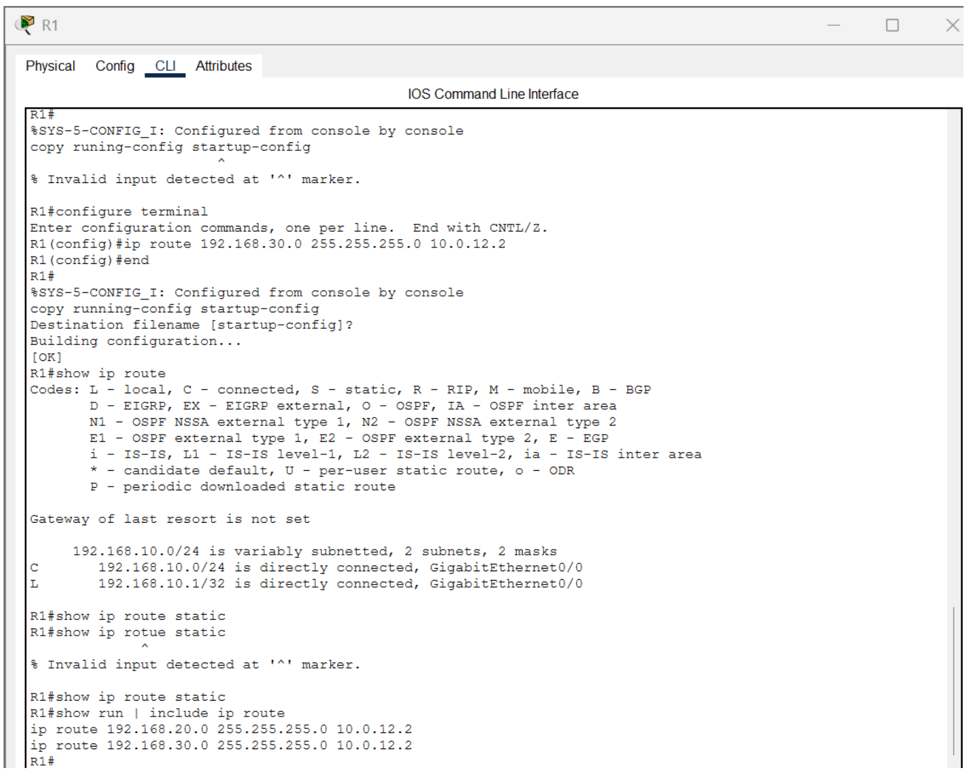
The /30 masks on the router-to-router links were chosen deliberately: each link only ever needs two usable host addresses (one per router interface), so a /30 avoids wasting a full /24 worth of address space on a link that will never have more than two devices on it.

4. Router Configuration

Each router was configured with its interface addresses and static routes toward the two LANs it does not directly serve. The core commands used on every router followed this pattern:

```
interface gigabitEthernet 0/0
ip address <ip> <mask>
no shutdown

ip route <destination network> <mask> <next-hop>
```

R1 was given static routes to 192.168.20.0/24 and 192.168.30.0/24 via 10.0.12.2. R2 was given routes to 192.168.10.0/24 via 10.0.12.1 and 192.168.30.0/24 via 10.0.23.2. R3 was given routes to 192.168.10.0/24 and 192.168.20.0/24 via 10.0.23.1. Every configuration was saved with copy running-config startup-config.




```
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy runing-config startup-config
                      ^
% Invalid input detected at '^' marker.

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 192.168.30.0 255.255.255.0 10.0.12.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0

R1#show ip route static
R1#show ip rotue static
             ^
% Invalid input detected at '^' marker.

R1#show ip route static
R1#show run | include ip route
ip route 192.168.20.0 255.255.255.0 10.0.12.2
ip route 192.168.30.0 255.255.255.0 10.0.12.2
R1#
```

Figure 2 — R1 CLI: static route added, saved to startup-config, and confirmed with show ip route (192.168.10.0/24 shown as directly connected).

5. Connectivity Testing

5.1 Interface Verification

```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     10.0.12.1       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>
```

Figure 3 — show ip interface brief confirming all configured interfaces are up/up.

5.2 Ping Testing

PC1 was used to ping across all three LANs. All pings completed successfully with 0% packet loss, confirming end-to-end connectivity through both routers.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=128
Reply from 192.168.30.10: bytes=32 time=19ms TTL=128
Reply from 192.168.30.10: bytes=32 time=20ms TTL=128
Reply from 192.168.30.10: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 10ms

C:\>
```

Figure 4 — Successful ping from PC1 to 192.168.30.10 (0% loss, TTL 128, replies of 1–20 ms).

5.3 Traceroute

A traceroute from PC1 to PC3 (192.168.30.10) was performed to observe the path taken by the packets hop by hop.

```
C:\>tracert 192.168.30.10

Tracing route to 192.168.30.10 over a maximum of 30 hops:

  1   0 ms      0 ms      1 ms      192.168.10.1
  2   *         *         *         Request timed out.
  3   *         *
```

Figure 5 — Traceroute output: hop 1 (192.168.10.1 / R1) responds; the capture was taken before later hops finished resolving.

The first hop confirms that traffic from PC1 leaves the local LAN through its default gateway, R1, exactly as expected — a directly connected device would show a response at hop 1 with the destination's own address, not an intermediate router. From here, each subsequent router (R2, then R3) would decrement the TTL by one and reply in turn, which is how traceroute maps out the full path PC1 → R1 → R2 → R3 → PC3 in a completed run. The key point this hop confirms either way: the packet does not travel directly from PC1 to PC3, because the two are on different IP networks. PC1 has no Layer 2 path to PC3, so it forwards every off-subnet packet to its default gateway, which then makes an independent Layer-3 forwarding decision at each hop based on its own routing table.

6. Traffic Analysis

6.1 ARP

Before PC1 could send its first ICMP echo request to a device outside its own LAN, it needed to resolve the MAC address of its default gateway (R1). Because PC1 did not yet have this mapping cached, it broadcast an ARP request onto LAN 1. R1 replied with its own MAC address, which PC1 then cached and used to address the Ethernet frame carrying the ICMP request. This ARP exchange was captured and stepped through in Packet Tracer's Simulation Mode before the ICMP traffic was observed, confirming that Ethernet frames rely on ARP-resolved MAC addresses even though the ping itself is addressed using IP.

6.2 ICMP

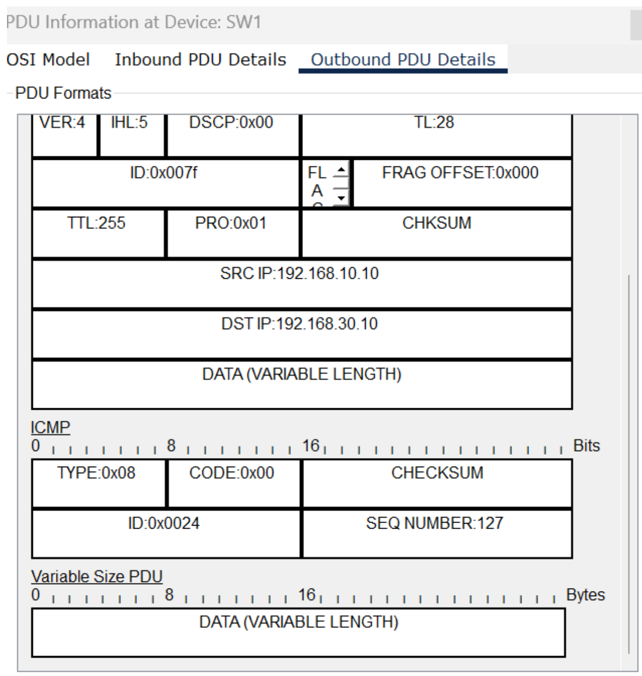


Figure 6 — Outbound PDU details at SW1 for an ICMP Echo Request: source IP 192.168.10.10, destination IP 192.168.30.10, TTL 255, ICMP Type 0x08 (Echo Request).

This capture shows the IP header and ICMP header of the first ping generated from PC1 toward PC3. The source IP (192.168.10.10) and destination IP (192.168.30.10) stay identical at every hop of the journey — this is the whole point of Layer 3 addressing, it identifies the true end-to-end endpoints. What does change at each hop is the Layer 2 (Ethernet) framing: every time the packet crosses a router, the router strips the old source/destination MAC addresses and re-encapsulates the packet with new MAC addresses appropriate for the next physical segment. TTL also decrements by one at every router hop; this is what allows tools like traceroute to work, and it is also what prevents a misconfigured or looping route from circulating a packet forever.

6.3 TCP/UDP

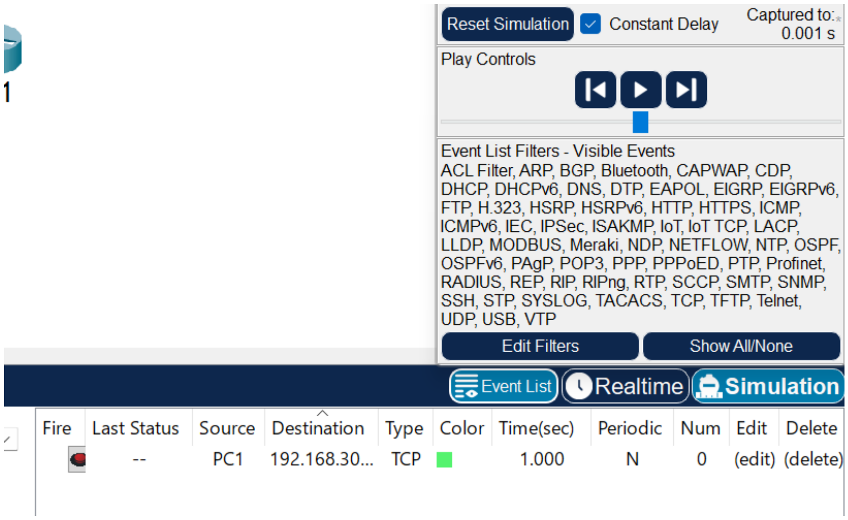


Figure 7 — Simulation Mode event list showing a TCP segment generated from PC1 toward 192.168.30.x, alongside the full list of protocols Packet Tracer can filter and display.

Generating TCP traffic in Simulation Mode made it possible to compare it against the ICMP/ARP traffic already observed. TCP traffic carries source and destination port numbers and establishes a connection (the three-way handshake) before any application data is exchanged, and it guarantees delivery and ordering through acknowledgements and retransmission. UDP, in contrast, has no handshake and no delivery guarantee — it simply sends datagrams and lets the application layer deal with any loss. This connection-oriented vs. connectionless distinction is why protocols like DNS queries or streaming use UDP for speed, while protocols like HTTP, SSH, and file transfers use TCP for reliability.

From a security-analyst perspective, source and destination ports matter because they identify which service is actually being talked to (e.g., port 443 for HTTPS, port 22 for SSH). An unusual or unexpected destination port on a connection — for example, an internal host suddenly talking to an external IP on port 4444 — is a classic indicator of a reverse shell or command-and-control channel, and is often one of the first things a SOC analyst checks. Protocol identification matters for the same reason: knowing whether traffic is TCP or UDP, and which port it is using, is what turns a stream of raw packets into a meaningful story about what a host is actually doing on the network. Network defenders need to understand both Layer 3 (where traffic is going) and Layer 4 (what it is doing when it gets there), because either one alone gives an incomplete picture — an IP address alone doesn't tell you if a connection is a normal web request or a suspicious backdoor.

7. Security Analysis

Observation 1 — ARP

ARP works by broadcasting a request ('who has this IP?') to every device on the local segment and accepting whichever reply comes back, with no authentication built in. This is exactly its main weakness: an attacker on the same LAN can send forged ('spoofed') ARP replies claiming to own an IP address they don't, redirecting traffic meant for the legitimate device to themselves — this is the basis of ARP spoofing / ARP poisoning, commonly used to perform man-in-the-middle attacks on a local network.

Observation 2 — ICMP

ICMP is extremely useful for troubleshooting because it gives immediate, low-overhead feedback about reachability (ping) and path (traceroute), without needing an application-layer connection at all. At the same time, excessive or unexpected ICMP traffic is worth investigating from a security standpoint: a sudden burst of ICMP echo requests can indicate a ping sweep (an attacker scanning a network to find live hosts), and unusually large or frequent ICMP packets can indicate an attempted ICMP flood (a denial-of-service technique) or even data exfiltration hidden inside ICMP payloads.

Observation 3 — MAC Addresses

MAC addresses change at every hop because they only have meaning on the local physical segment they were assigned for. When a router receives a frame, it discards the incoming Layer 2 header entirely, looks only at the Layer 3 (IP) destination to decide where the packet should go next, and then builds a brand-new Ethernet frame addressed to whichever device is next in line (the next router, or the final destination). The IP addresses inside stay the same throughout the journey because they identify the true source and destination end to end, while MAC addresses are only ever a 'next hop' address.

Observation 4 — Ports

Source and destination ports identify which application or service is generating or receiving traffic on a host that may be running many services at once. A destination port of 80 or 443 usually points to a web server, 22 to SSH, 53 to DNS, and so on. Seeing consistent, well-known ports in traffic is a sign of normal activity, while high or unregistered ports being used unexpectedly (especially as a destination) can flag traffic that deserves closer investigation.

Observation 5 — Routing

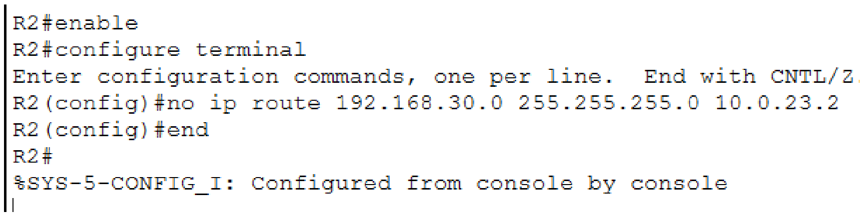
If R2 did not have a route to 192.168.30.0/24, then any packet arriving at R2 destined for that network would have no matching entry in R2's routing table. R2 would be unable to forward it, and — assuming no default route existed either — R2 would drop the packet and, in a real Cisco device, typically respond to the sender with an ICMP 'Destination Unreachable' message. This is essentially what was demonstrated directly in the troubleshooting exercise below.

8. Troubleshooting Exercise

To practice diagnosing a real network fault, the static route to 192.168.30.0/24 was intentionally removed from R2 (Option A from the assignment), simulating a misconfiguration that a network engineer might encounter in production.

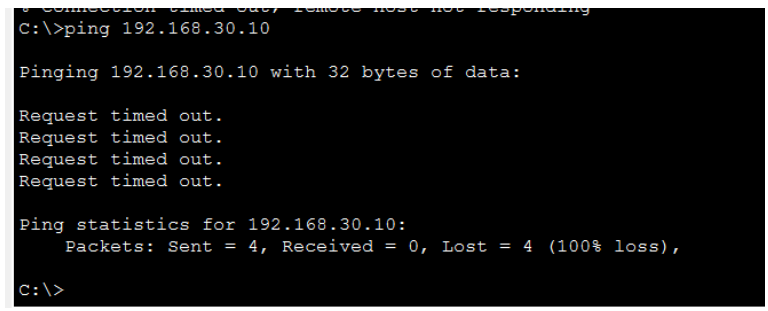
Stage	Detail
Problem	The static route "ip route 192.168.30.0 255.255.255.0 10.0.23.2" was manually removed from R2 using "no ip route ...".
Evidence	PC1 could no longer ping 192.168.30.10; the ping timed out with "Request timed out" for all four attempts.
Diagnosis	"show ip route" was run on R2 and compared against the earlier working output; the 192.168.30.0/24 entry (marked S) was confirmed missing.
Solution	The route was re-added on R2: ip route 192.168.30.0 255.255.255.0 10.0.23.2, then saved

Stage	Detail
	with copy running-config startup-config.
Verification	Ping from PC1 to 192.168.30.10 was re-run and succeeded with 0% packet loss, confirming the network was restored.



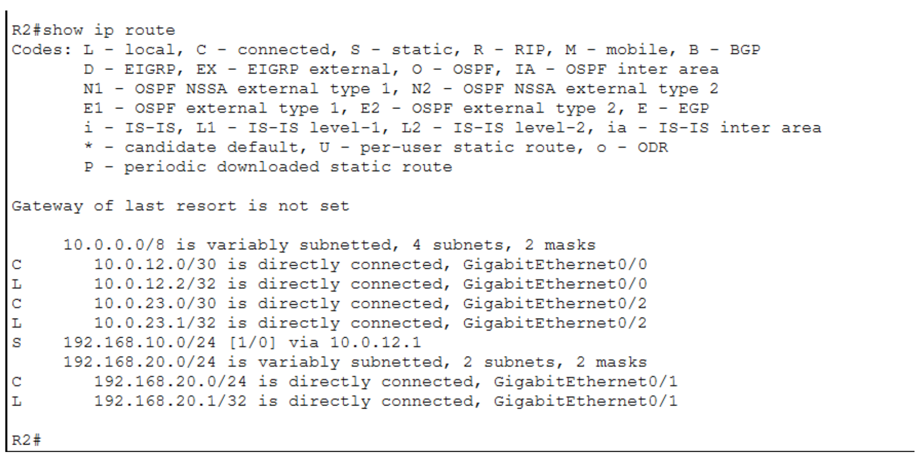
```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no ip route 192.168.30.0 255.255.255.0 10.0.23.2
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 8 — Static route to 192.168.30.0/24 removed from R2.



```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 9 — Ping from PC1 to PC3 failing after the route was removed.



```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/0
L       10.0.12.2/32 is directly connected, GigabitEthernet0/0
C       10.0.23.0/30 is directly connected, GigabitEthernet0/2
L       10.0.23.1/32 is directly connected, GigabitEthernet0/2
S    192.168.10.0/24 [1/0] via 10.0.12.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1

R2#
```

Figure 10 — show ip route on R2 confirming the 192.168.30.0/24 route is missing (diagnosis step).

9. Required Online Learning Labs

All required online labs and the networking CTF challenge were completed in full, as evidenced below.

Lab	Platform	Status
Network Packet Analysis	LetsDefend	Completed

Lab	Platform	Status
Network Services	TryHackMe	Completed
Network Services 2	TryHackMe	Completed
DNS in Detail	TryHackMe	Completed
Network Security Essentials	TryHackMe	Completed
Pickle Rick (Challenge)	TryHackMe	Completed

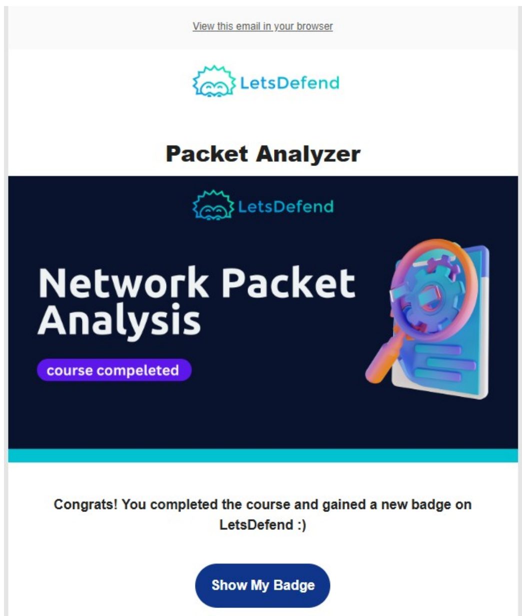


Figure 11 — LetsDefend "Network Packet Analysis" course completion badge.

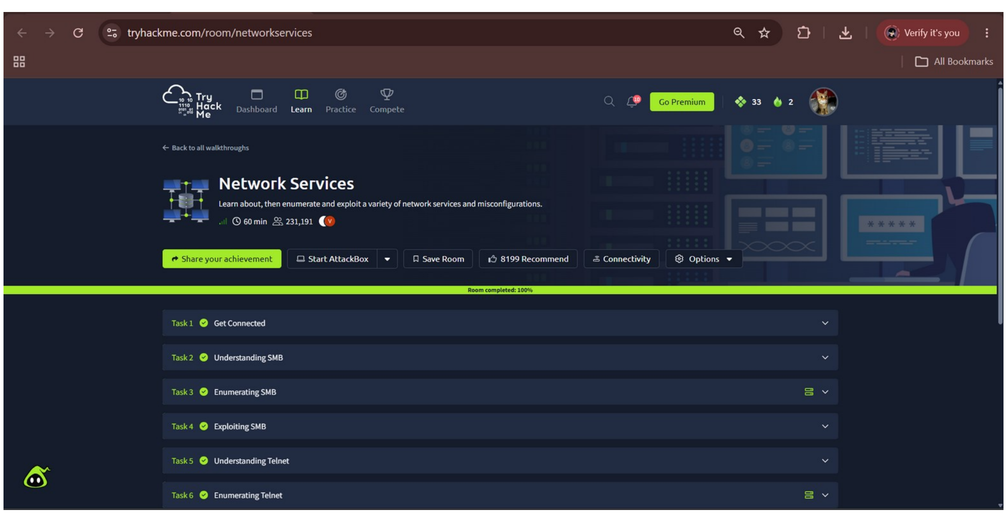


Figure 12 — TryHackMe "Network Services" room, 100% complete.

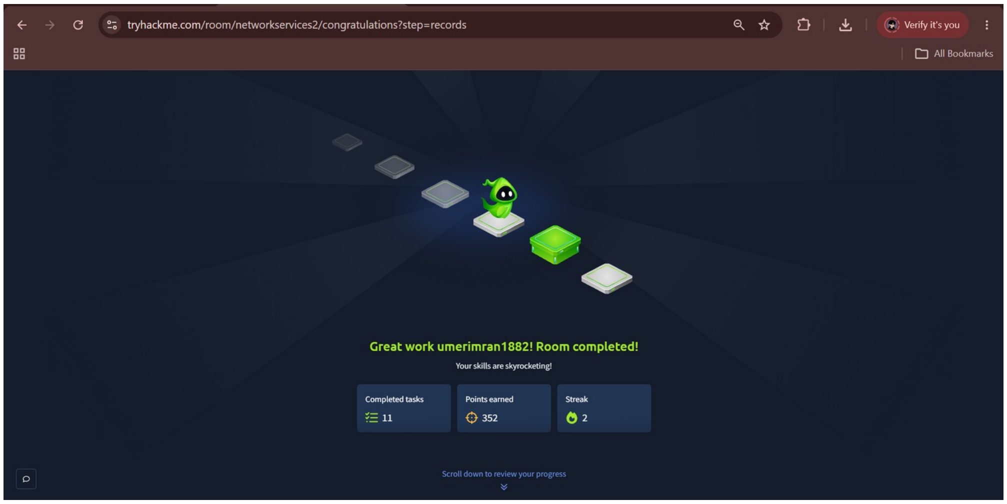


Figure 13 — TryHackMe "Network Services 2" room, 100% complete.

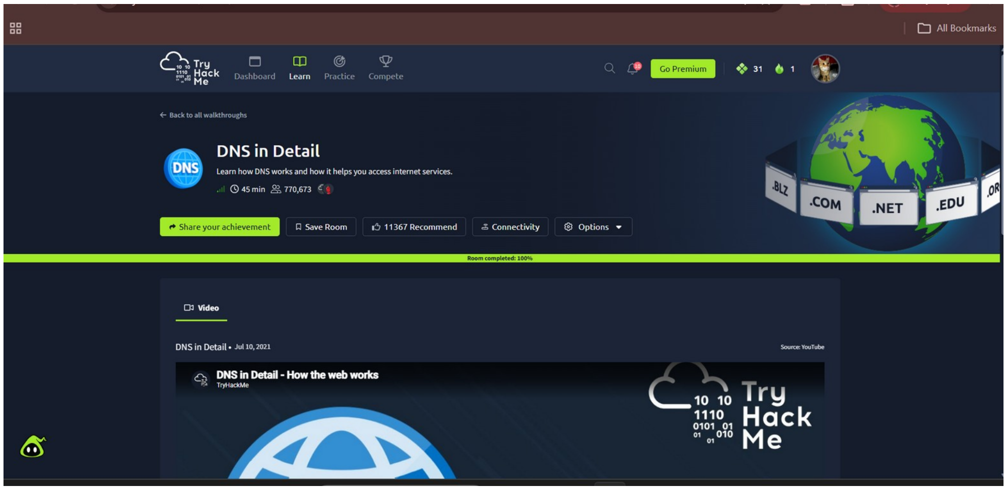


Figure 14 — TryHackMe "DNS in Detail" room evidence.

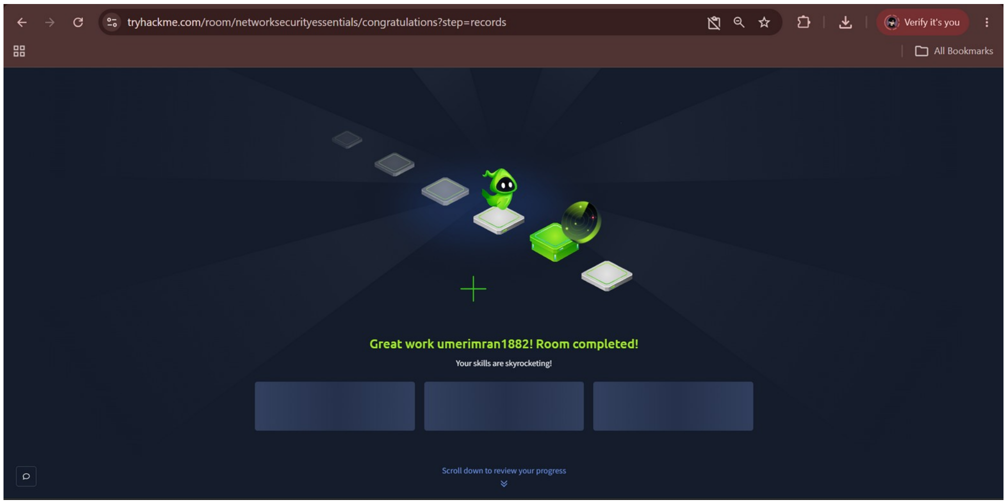


Figure 15 — TryHackMe "Network Security Essentials" room, completed.

Pickle Rick — Networking Challenge

The Pickle Rick room was approached as a practical challenge rather than a walkthrough. Reconnaissance began with an Nmap scan of the target to identify open ports and running services, which revealed a web server. Browsing the site and checking robots.txt led to further hidden paths, and directory brute-forcing (Gobuster) was used to enumerate additional content that was not linked from the visible site. This led to a login/command panel that allowed command execution once valid credentials were located, which was then used to explore the filesystem and locate the three hidden 'ingredients' the room is built around.

Networking concepts that were directly relevant here included understanding how a web application exposes ports and services that can be enumerated (tying back to the TCP/port discussion earlier in this report), how HTTP requests and hidden endpoints work, and how a service running on a non-obvious path is still discoverable through methodical reconnaissance rather than guesswork — the same mindset used earlier when tracing ARP, ICMP, and TCP traffic through the Packet Tracer topology.

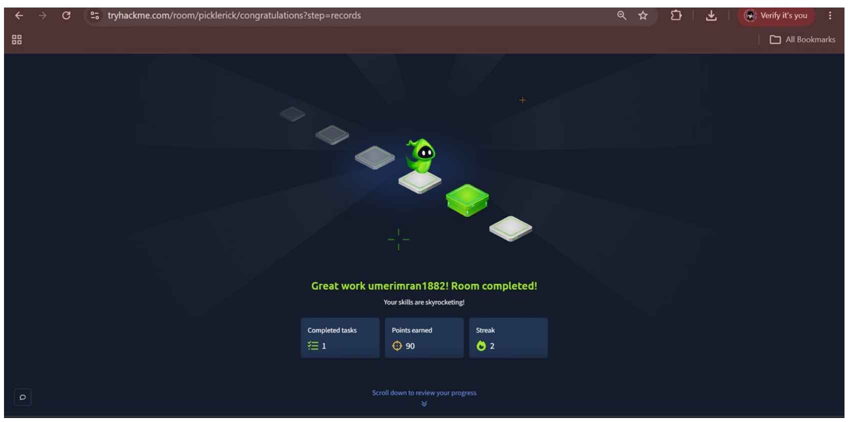


Figure 16 — TryHackMe Pickle Rick room marked as completed.

10. Assessment Questions

Networking Fundamentals

Q1. What is the difference between a switch and a router?

A switch works inside one network — it looks at MAC addresses and forwards frames between devices on the same LAN, like connecting PC1 to R1 through SW1. A router is what connects separate networks together. It looks at the IP address instead of the MAC address and decides which direction a packet needs to go next. Basically, a switch keeps traffic moving inside a network, and a router is what lets traffic leave one network and get to another — which is exactly what R1, R2 and R3 were doing for the three LANs in this lab.

Q2. Why does every PC require a default gateway to communicate with another network?

Because a PC can only talk directly to devices on its own subnet. If PC1 wants to send something to PC3, it checks the destination IP and realizes it's not on 192.168.10.0/24, so it has no direct way to deliver it. That's where the default gateway comes in — the PC just hands the packet to R1 and lets the router figure out the rest. Without a gateway configured, the PC would have no idea where to even send traffic that isn't local.

Q3. What is a subnet mask?

It's basically what tells a device where the "network part" of an IP address ends and the "host part" begins. So for 192.168.10.10 with mask 255.255.255.0, the device knows the first three octets identify the network and the last octet identifies the specific host. This is how a PC decides if a destination is local (send directly) or remote (send to the gateway).

Q4. Why were /30 networks used between the routers?

A /30 only gives you 2 usable addresses, and that's literally all a link between two routers needs — one address for each end. Using something bigger like a /24 on a link that will only ever have 2 devices on it would just waste a huge chunk of address space for nothing. This is actually one of the things that made more sense to me once I saw it applied here instead of just reading the definition.

Q5. What is the purpose of a routing table?

It's the list a router checks every single time it needs to forward a packet — basically "if the destination is this network, send it out this interface / to this next-hop." Without it the router has no way of knowing where anything outside its own directly connected interfaces should go. I could actually see this happening in Step 13, comparing the S entries on R1, R2 and R3 and matching them to the addressing table.

Q6. What is a static route?

It's a route that I typed in manually with the ip route command instead of the router learning it automatically through something like OSPF. It just tells the router "to reach this network, send traffic to this next-hop IP," and it stays exactly like that until someone changes it.

Q7. What are the advantages and disadvantages of static routing?

The good part is it's simple, predictable, and doesn't use up any extra bandwidth or CPU running a routing protocol in the background — for a small 3-router setup like this one it was honestly pretty quick to set up and easy to verify. The downside showed up during the troubleshooting step: the second I deleted one route, the whole network broke and nothing fixed itself. In a bigger network, or one where links go down often, that becomes a real problem because someone has to manually go and fix every static route by hand instead of the network adapting on its own like it would with a dynamic protocol.

Traffic Analysis

Q8. What is ARP?

ARP is the protocol that finds out which MAC address belongs to a certain IP address on the local network. PC1 knows R1's IP address (192.168.10.1) but it doesn't automatically know R1's MAC address, so it sends out an ARP request asking "who has this IP" and R1 answers back with its MAC.

Q9. What is ICMP?

ICMP is used for network diagnostics and error reporting rather than actually carrying application data. Ping uses ICMP Echo Request/Reply to test if a device is reachable, and it's also what generates the "Destination Unreachable" or "Time Exceeded" messages that traceroute relies on.

Q10. What is TTL?

TTL stands for Time to Live and it's a number in the IP header that goes down by 1 every time the packet passes through a router. Once it hits 0 the router just drops the packet instead of forwarding it. This exists mainly so a packet can't get stuck looping forever if there's ever a routing loop — it forces the packet to eventually die instead of circulating endlessly.

Q11. What happens to Ethernet frames when they cross a router?

They don't survive the crossing, basically. The router opens up the packet, throws away the old Ethernet header completely, and builds a new one with different source/destination MAC addresses before sending it out the next interface. This was actually one of the more confusing parts for me at first — I kept assuming the same frame just travels the whole way, but it's really the IP packet that survives the trip, not the frame around it.

Q12. Why does the IP source/destination remain logically end-to-end while MAC addresses are hop-by-hop?

Because they're doing two completely different jobs. The IP address is answering "who is this packet ultimately for," so it has to stay the same the whole way or the packet would lose track of where it's actually going. The MAC address is only answering "who is the very next device on this wire," so it only needs to make sense for one hop at a time and gets rebuilt at every router.

Q13. What information can a security analyst obtain from source/destination ports?

Ports basically tell you what kind of traffic you're looking at without even opening the packet contents. Port 443 → probably HTTPS, port 22 → SSH, port 53 → DNS, and so on. If an analyst sees a host talking to some random high-numbered port they don't recognize, especially to an external IP, that's usually a red flag worth checking — could be malware calling home or something similar.

Q14. What was the most interesting packet you observed and why?

Honestly it was the ICMP packet in Figure 6. It doesn't sound exciting written down, but seeing the actual TTL value and both the source/destination IP sitting right there in the PDU view made something click that reading about it never did — like I could literally watch the same IP addresses stay put while I already knew the MAC addresses were about to get swapped at the next hop. Small thing, but it was the moment this stopped feeling like theory.

Security

Q15. Identify two security concerns related to network traffic observed in this lab.

First one is ARP — since there's no authentication on ARP replies, anyone on the same LAN could send a fake reply and trick devices into sending traffic to them instead of the real gateway. Second one is ICMP — it's so lightweight and unauthenticated that it's an easy way for someone to scan a network (ping sweep) just to find out which hosts are alive before attacking anything further.

Q16. Explain one potential attack involving ARP.

ARP spoofing is probably the classic example. An attacker sends out fake ARP replies saying "this IP belongs to my MAC address" for something like the default gateway. Devices on the LAN just trust it and update their ARP cache, so now their traffic that's supposed to go to the router actually goes to the attacker first. The attacker can read it, change it, or just forward it along so nobody notices — that's basically a man-in-the-middle attack, and it works because ARP has no way to check if a reply is actually legit.

Q17. Explain why network traffic analysis is useful for incident detection.

Because you can't always trust a host to tell you honestly what it's doing, especially if it's already been compromised. Looking at the actual traffic instead — where it's going, what ports, how often — gives you a way to catch stuff a compromised machine would never report about itself, like a connection going out to some server it has no business talking to.

Q18. Explain how understanding normal network traffic helps identify abnormal traffic.

You basically need a baseline before "abnormal" even means anything. If I don't know what normal ARP behavior or normal port usage looks like on this network, then a spoofed ARP reply or a weird destination port just blends in with everything else. But once you've seen what's normal — like I did comparing R2's routing table before and after I broke it — anything that doesn't match stands out pretty fast.

11. Reflection

Going into this week I thought it would mostly be about getting the ping to work between the three PCs, but the part that actually taught me something was switching into Simulation Mode after everything was already working. Watching PC1 send out an ARP request before it could even send the first ping, and then watching the same ICMP packet keep its IP addresses the whole way while the MAC addresses got swapped out at every router — that's the kind of thing that's easy to read about but doesn't really sink in until you watch it happen packet by packet.

The troubleshooting part was honestly the hardest bit, not because deleting a route is technically hard, but because I had to actually diagnose it properly instead of just guessing and re-adding things until it worked. Comparing R2's routing table before and after made me realize how much of security monitoring is really just "know what normal looks like so you can spot what isn't." What surprised me most was how much you can tell from one single ICMP packet — TTL, both headers, the works — once you actually know what you're looking at.

Overall this made it pretty clear that networking isn't really a separate topic from cybersecurity, it's more like the foundation everything else sits on top of. If I don't understand what normal traffic looks like, I have no shot at recognizing abnormal traffic later on. Next I want to try this with an actual Wireshark capture instead of a simulator, and maybe look into what ARP spoofing actually looks like on a real packet capture rather than just reading about it.